

Quantum Career Roadmaps

By Natashaia Kaur Raina

About Myself

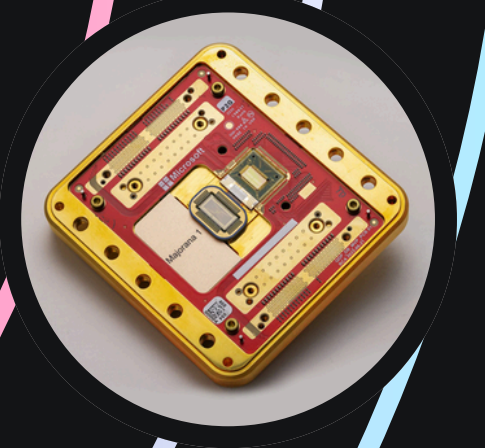
- Editor in Chief at STEAMI covering STEM News
- Building an early-stage Quantum Tech Product for an upcoming Startup
- Building my Quantum EdTech Startup
- Classiq Brand Ambassador
- Quantum Tech Youtuber and Podcast Host
- Completed Quantum Research Internships in Quantum Communication and QML
- Undergraduate Student at IIT Madras



What We'll Cover Today



- **Introduction to Quantum Computing**
- **Why Quantum Computing**
- **India and Quantum Tech**
- **Global Landscape**
- **Quantum Learning Roadmap**
- **Quantum Career Pathways**
- **Emerging Quantum Fields**
- **Why Join Quantum**
- **Who Should Join Quantum**
- **Q&A Session**



Bits

- A bit is the smallest unit of data in computing.
- It can have only two possible values: 0 or 1.
- 0 usually represents OFF, false, or low voltage, and 1 represents ON, true, or high voltage.
- Multiple bits can be combined to represent more complex data:
- 8 bits = 1 byte
- 16 bits = 2 bytes
- Forming larger data structures (like integers or floating-point numbers).
- All digital data (text, images, videos, etc.) is ultimately represented using bits.

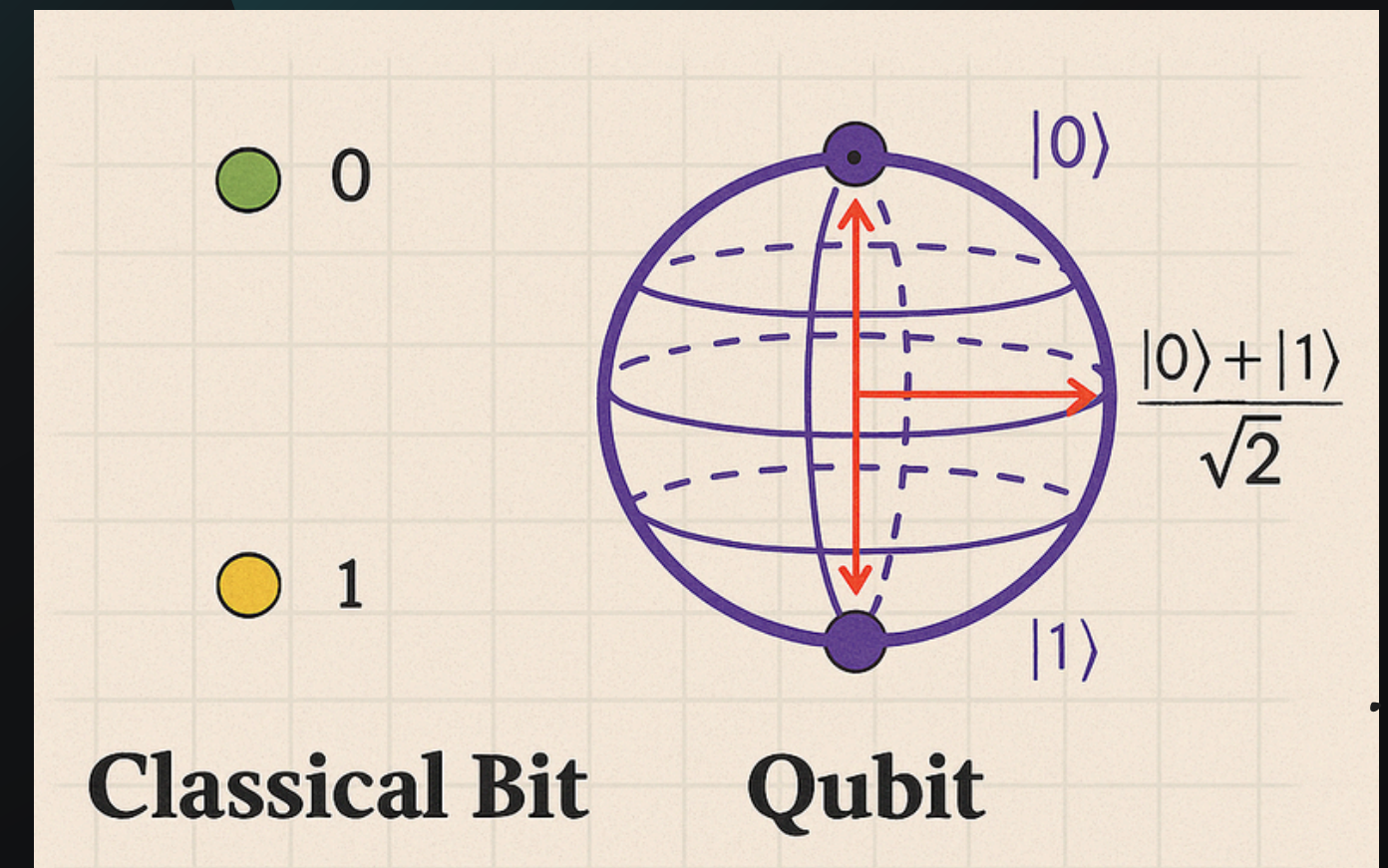
BIT BINARY DIGIT



**A BIT HAS A VALUE
OF EITHER 0 OR 1.**

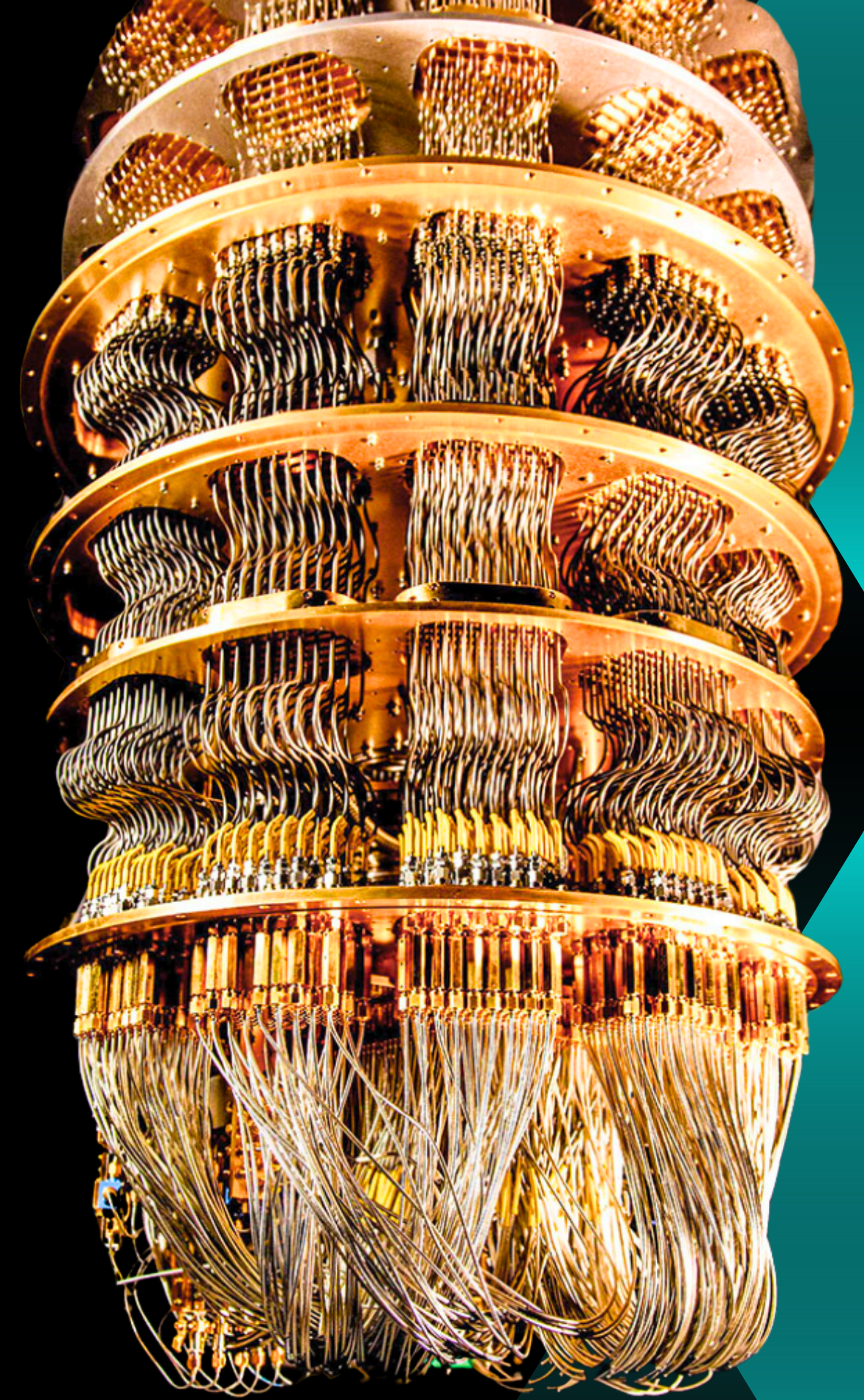
Qubits

- A qubit is the basic unit of quantum information
- Unlike a classical bit, a qubit can exist in a superposition of both $|0\rangle$ and $|1\rangle$ states simultaneously.
- A qubit's state is described by:
- $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ where α and β are complex probability amplitudes satisfying $|\alpha|^2 + |\beta|^2 = 1$.
- Measurement collapses a qubit's superposition to either 0 or 1, with probabilities determined by $|\alpha|^2$ and $|\beta|^2$.
- The squared magnitudes of these amplitudes give the probabilities of measuring the qubit in state $|0\rangle$ or $|1\rangle$ states.
- Qubits can be implemented using various physical systems: trapped ions, superconducting circuits, photons, etc.



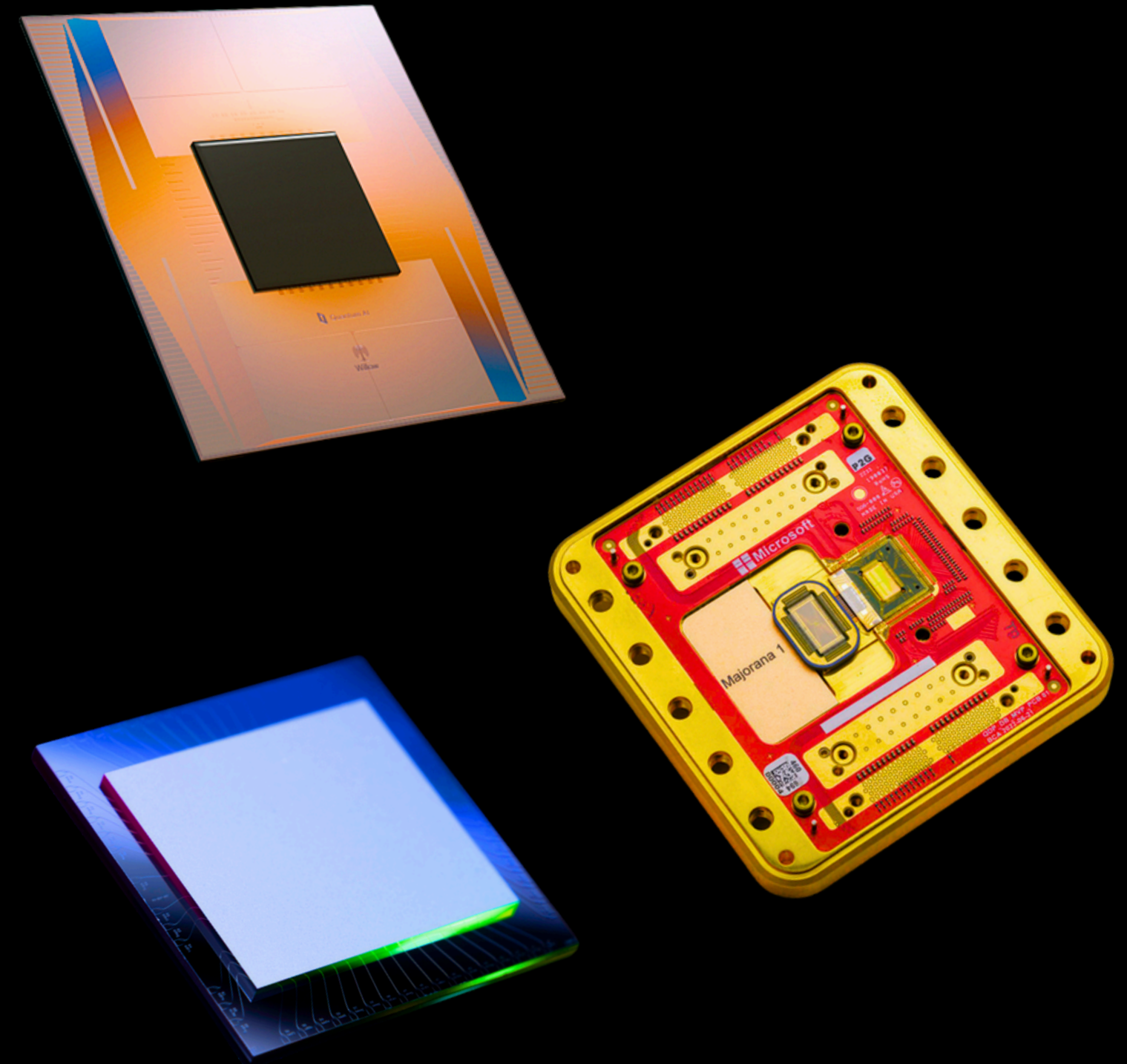
Quantum Computers

- Quantum computers use qubits and quantum effects to solve problems classical computers can't.
- They operate at ultra-cold temperatures using dilution refrigerators to preserve quantum states.
- They often look like golden chandeliers, housing the quantum processor and control systems.



Quantum Chips

- Quantum chips contain qubits and quantum circuits for performing quantum computations.
- They are made using platforms like superconducting circuits, trapped ions, or photonics.
- Operate at cryogenic temperatures to maintain coherence and reduce noise.



Types of Quantum Computers

Superconducting Qubits

Use circuits cooled to near absolute zero to exhibit quantum behavior.

Fast gate operations and widely used by IBM, Google, and Rigetti.

Neutral Atoms

Qubits are individual neutral atoms trapped using optical tweezers.

Highly scalable and reconfigurable; used by companies like QuEra and PASQAL.

Photonic

Use single photons as qubits, manipulated via beam splitters and phase shifters.

Room-temperature operation and good for quantum communication tasks.

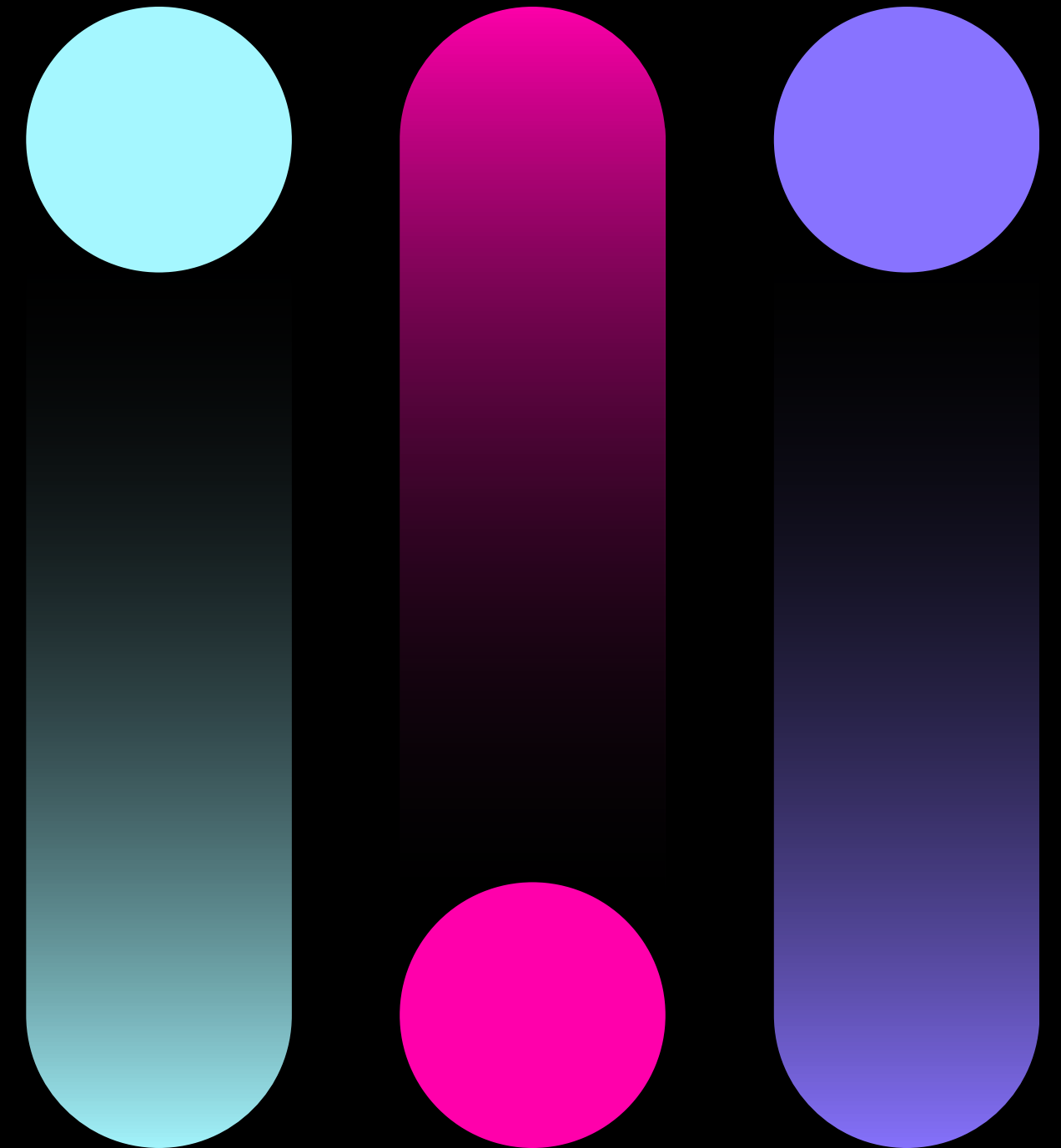
Trapped-Ion

Qubits are individual ions trapped and manipulated using lasers.

High fidelity and long coherence times, used by IonQ and Quantinuum.

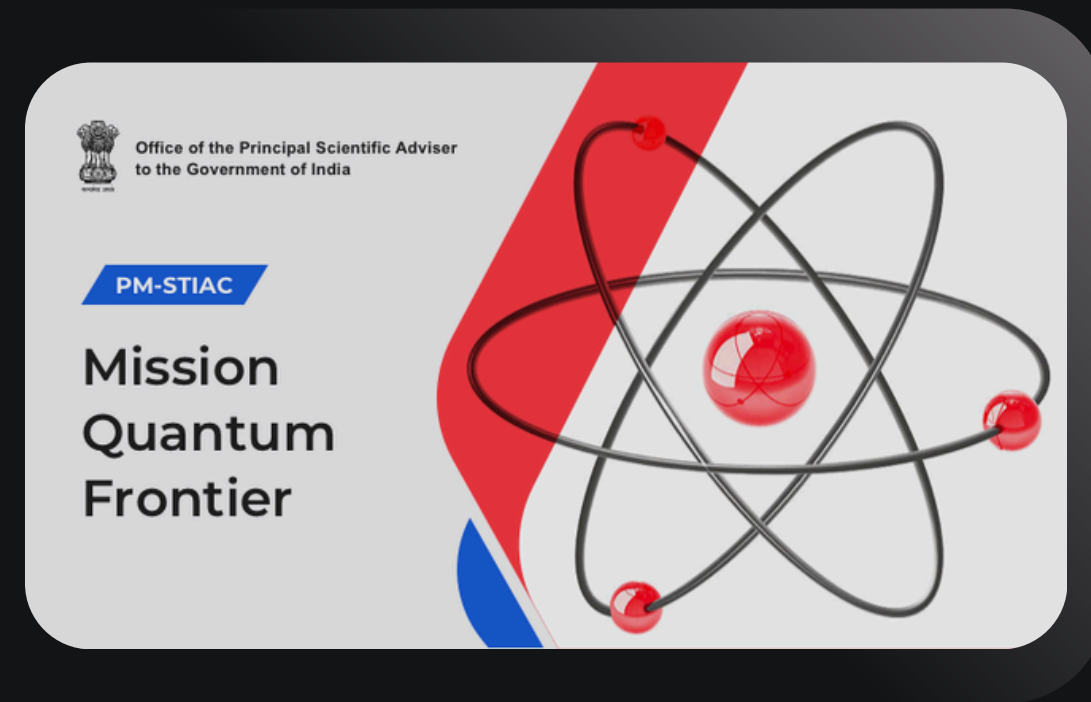
Why Quantum Computing?

- Transistors are the fundamental building blocks of modern computers they act as tiny on/off switches that control the flow of electrical current.
- Traditionally, they follow the rules of classical physics, where electrons behave like tiny, localized particles.
- But as we shrink transistors to just a few nanometers (1 nanometer = 1 billionth of a meter), we're approaching the scale of individual atoms and at this point, quantum mechanics kicks in.
- There are limitations to classical computing but quantum computing hold the potential to exponentially increase computational power.



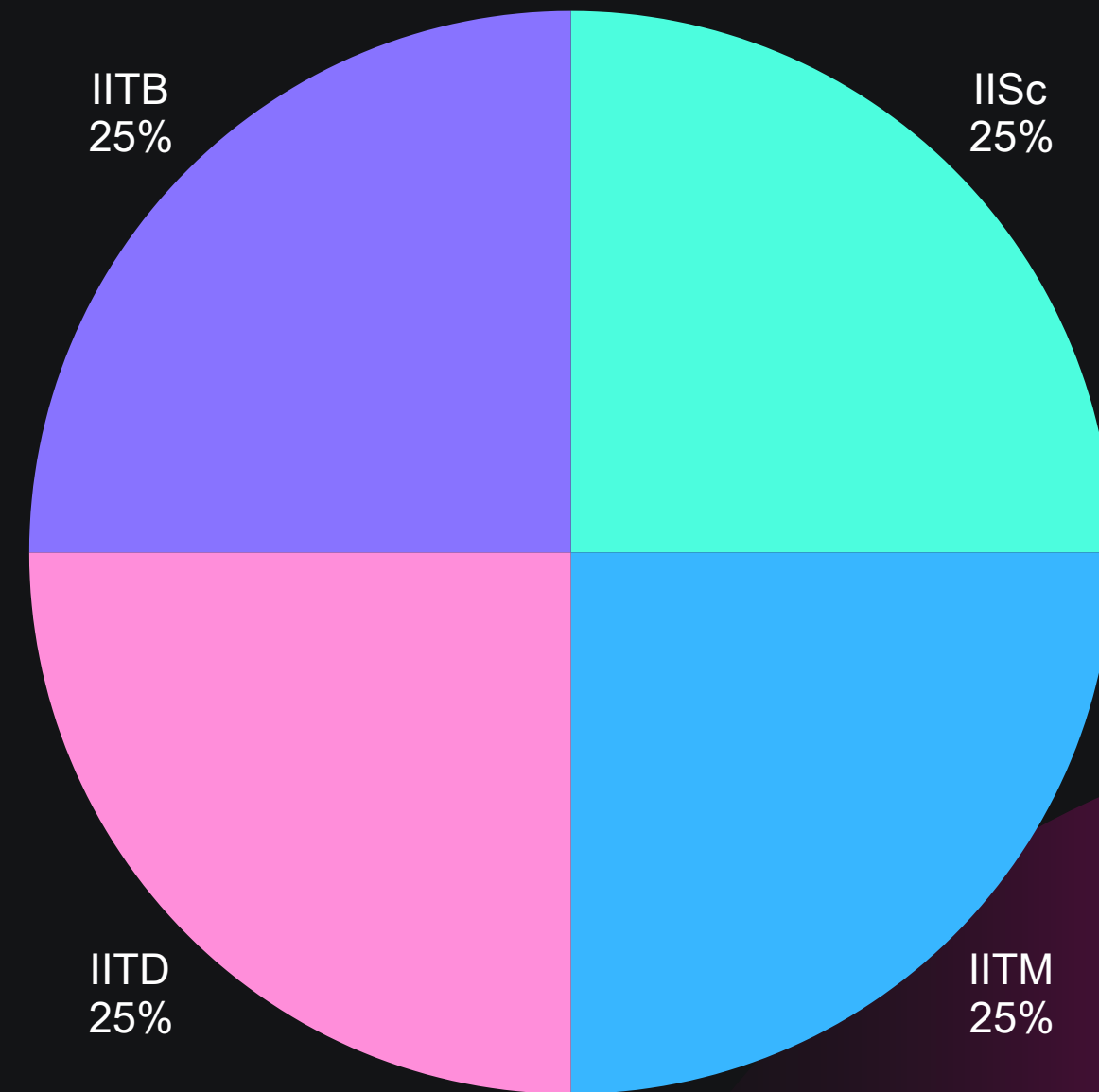
National Quantum Mission(NQM)

- Quantum Valley
- 4 Thematic Hubs
- Undergraduate Minor Course
- BTech in Quantum Computing
- ₹6,003.65 crore (2023–2031)
- 8 NQM backed quantum startups
- IIT Delhi (with DRDO support): 1 km free-space entanglement based secure link demo
- Other research, startup, education schemes to push quantum in India



4 Thematic Hubs

1. IISc - Quantum Computing
2. IIT Madras - Quantum Communication
3. IIT Bombay - Quantum Sensing & Metrology
4. IIT Delhi - Quantum Materials & Devices



Quantum Learning Roadmap

Step 1: Master Math Fundamentals (Linear Algebra, Complex Numbers, Probability)

Step 2: Learn the Basics of Quantum Mechanics (Superposition, Entanglement, Measurement)

Step 3: Pick up Python + Start Using a QSDK (Qiskit, PennyLane, Cirq, etc.)

Step 4: Implement Key Algorithms (Grover's, Shor's, Quantum Fourier Transform)

Step 5: Dive into Advanced Fields like Quantum Machine Learning & Quantum Cryptography

Step 6: Build Projects, Join Hackathons, Internships & Quantum Communities

Emerging Quantum Fields

1. Quantum Machine Learning (QML)

Accelerates AI by improving pattern recognition, deep learning, and generative models. Applications in finance, healthcare, and scientific research.

2. Quantum Cryptography

Future-proofing security with Quantum Key Distribution (QKD) and post-quantum cryptography. Helps protect financial transactions, government data, and global communications.

3. Quantum Networking

Developing the foundations for a Quantum Internet, enabling secure, ultra-fast communication. Potential use cases in cloud computing, secure messaging, and distributed quantum computing.

Emerging Quantum Fields

4. Quantum Simulation

Simulating complex molecules and materials that classical computers struggle with. Key for advancements in drug discovery, material science, and chemistry.

5. Quantum Optimization

Enhancing efficiency in logistics, financial modeling, and resource allocation. Could improve supply chain management, transportation, and investment strategies.

6. Quantum Biology & Energy

Exploring quantum effects in photosynthesis, enzyme activity, and molecular interactions. Potential breakthroughs in clean energy, sustainable materials, and biotechnology.

Quantum Career Pathways



Quantum Researcher

Academia & labs (PhD → Faculty)

Industry R&D (IBM, Google, etc.)

Focus: algorithms, hardware, theory



Quantum Software Engineer

Build quantum circuits, simulators, SDKs (Qiskit, Cirq, etc.)



Quantum Machine Learning Engineer

Apply quantum computing to ML problems (QML, VQA, QNNs)

Work on hybrid quantum-classical models for AI, finance, chemistry

Quantum Career Pathways

Quantum Algorithms Developer

Design quantum algorithms for optimization, ML, cryptography, finance
Bridge theory + SWE

Quantum Cryptography Specialist

Work on QKD, post-quantum encryption
Startups like QNu Labs, ID Quantique

Quantum Product Manager / Consultant

Drive strategy, roadmap, and customer solutions
Need hybrid expertise: tech + business

Quantum Educator / Communicator

Build quantum literacy via content, courses, podcasts, media

1. Quantum Algorithm Developer

Key Responsibilities

- Researching and developing novel quantum algorithms that solve specific problems more efficiently than classical methods.
- Collaborating with quantum software engineers to implement algorithms on existing hardware or simulators.
- Testing and benchmarking algorithms, comparing performance against classical baselines.
- Publishing findings in academic journals and presenting at industry conferences.
- Keeping abreast of the latest hardware improvements and adjusting algorithmic approaches accordingly.

Essential Skills

- Mathematics & Quantum Mechanics: Deep familiarity with linear algebra, probability theory, and quantum operators.
- Programming: Proficiency in Python, C++, or MATLAB for simulating and testing algorithms.
- Quantum SDKs: Experience with toolkits like Qiskit (IBM), Cirq (Google), or Ocean (D-Wave).
- Problem-Solving: Ability to translate real-world challenges into quantum-friendly formulations.
- Research Acumen: Comfort interpreting complex research papers, running experiments, and exploring theoretical frontiers.

2. Quantum Software Engineer

Key Responsibilities

- Designing, coding, and maintaining quantum software libraries and development kits.
- Ensuring interoperability with cloud-based quantum services (e.g., AWS Braket, Microsoft Azure Quantum).
- Optimising compilation and circuit transpilation to accommodate hardware constraints (gate counts, error rates).
- Collaborating with hardware teams to implement calibration routines and error mitigation techniques.
- Streamlining development pipelines—continuous integration, testing, and deployment—for quantum applications.

Essential Skills

- Classical Software Engineering: Strong grasp of data structures, algorithms, and design patterns.
- Quantum Programming: Familiarity with quantum circuit libraries (Qiskit, Cirq, PyQuil) and domain-specific languages (e.g., Q#).
- CI/CD & DevOps: Experience in setting up automated testing, version control, and containerisation for quantum projects.
- Cross-Platform Knowledge: Understanding of differences among various quantum hardware backends.
- Collaboration: Working closely with quantum algorithm researchers and hardware teams to align software capabilities with evolving hardware needs.

3. Quantum Hardware Engineer

Key Responsibilities

- **Designing and fabricating qubits, resonators, and quantum chips in cleanroom environments.**
- **Managing cryogenic systems that maintain the ultra-low temperatures required for superconducting qubits.**
- **Developing control electronics and microwave systems to manipulate qubits and measure outputs.**
- **Analysing decoherence sources and implementing methods to extend qubit lifetimes.**
- **Collaborating with software teams to provide real-time feedback on hardware performance metrics.**

Essential Skills

- **Physics & Engineering:** Strong background in condensed matter physics, optics, or electronic engineering.
- **Lab Techniques:** Proficiency with cryogenics, vacuum technology, circuit design, and advanced measurement tools.
- **Material Science:** Knowledge of superconductors, semiconductor fabrication, or ion-trapping techniques.
- **Data Analysis:** Handling and interpreting large volumes of experimental data, often in Python or MATLAB.
- **Hands-On Problem-Solving:** Comfort troubleshooting intricate systems, from electronic noise issues to unexpected qubit decoherence.

4. Quantum Research Scientist (Theoretical/Experimental)

Key Responsibilities

- Conducting theoretical analyses of quantum systems, qubit interactions, and potential computational advantages.
- Running lab experiments to test hypotheses, gather data, and refine quantum hardware designs.
- Publishing research in peer-reviewed journals, attending conferences, and collaborating on multi-institutional projects.
- Proposing and securing grant funding to expand the organisation's research footprint.
- Mentoring junior team members, postdoctoral fellows, or PhD students.

Essential Skills

- **Quantum Mechanics:** In-depth understanding of entanglement, quantum gates, error correction protocols, and advanced quantum theory.
- **Research Methods:** Experience designing experiments, handling complex instrumentation, and conducting rigorous data analysis.
- **Mathematics & Programming:** Strong competence in linear algebra, differential equations, and computational modelling.
- **Critical Thinking:** Ability to question assumptions, evaluate data objectively, and pivot research directions when necessary.
- **Academic Excellence:** Track record of publishing and presenting in reputable platforms, indicating peer recognition.

5. Quantum Error Correction Specialist

Key Responsibilities

- Designing and implementing advanced error-correcting codes that minimise decoherence and logical qubit failure.
- Creating fault-tolerant circuit architectures that incorporate redundancy without excessive resource overhead.
- Collaborating with hardware teams to match error correction protocols to specific qubit technologies (superconducting, ion traps, etc.).
- Evaluating performance trade-offs—resource overhead vs. achievable gate fidelities, etc.
- Publishing findings and guiding best practices for quantum hardware/software communities.

Essential Skills

- **Mathematical Rigour:** Mastery of linear algebra, group theory, and coding theory.
- **Quantum Architecture Knowledge:** Understanding the unique error profiles of different qubit implementations.
- **Simulation & Tooling:** Ability to run large-scale simulations of error correction schemes using frameworks like Qiskit Ignis.
- **Algorithmic Thinking:** Designing solutions that balance theoretical correctness with real-world feasibility.
- **Interdisciplinary Collaboration:** Engaging with both theoretical physicists and hardware engineers to refine protocols.

6. Quantum Cryptography Engineer

Key Responsibilities

- Developing and implementing post-quantum cryptographic algorithms (lattice-based, code-based, hash-based, etc.).
- Integrating QKD systems into existing networks, ensuring secure key exchange and interoperability.
- Conducting vulnerability assessments of quantum and classical cryptographic protocols.
- Collaborating with standards bodies (NIST, ETSI) on defining post-quantum standards.
- Advising organisations on best practices for quantum-proofing systems, including hybrid encryption strategies.

Essential Skills

- **Cryptographic Foundations:** Understanding classical algorithms (RSA, AES) and post-quantum counterparts (CRYSTALS-Kyber, Dilithium).
- **Networking & Security:** Familiarity with secure communication protocols, key management, and PKI infrastructures.
- **Optics & Photonics (for QKD):** Knowledge of single-photon detectors, quantum channels, and photon polarisation.
- **Standards & Regulations:** Awareness of emerging global cryptographic standards and compliance requirements.
- **Implementation Skills:** Ability to code cryptographic libraries in efficient, secure ways (C/C++, Rust, or Go).

7. Quantum Machine Learning (QML) Researcher

Key Responsibilities

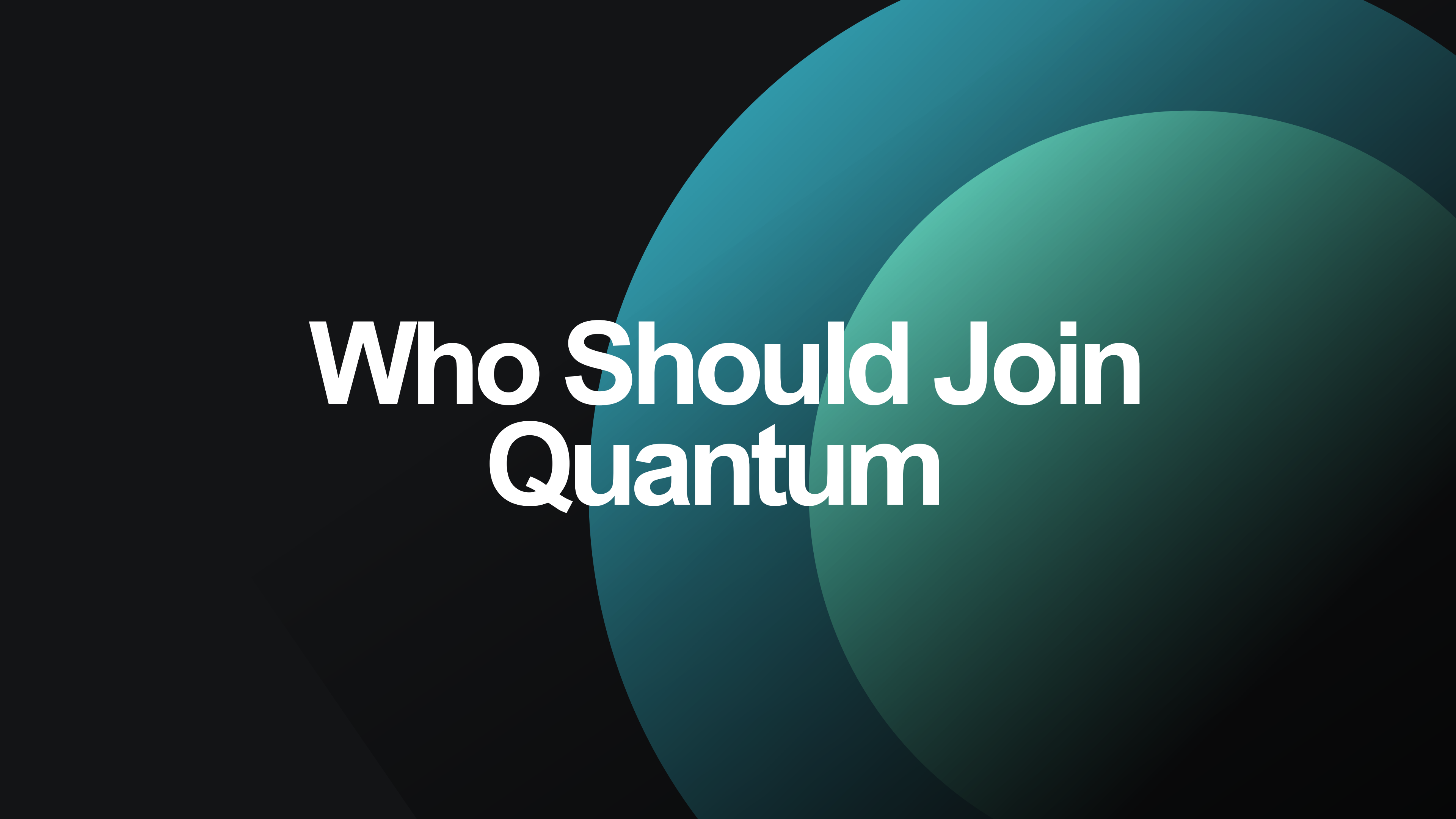
- Investigating quantum-accelerated algorithms for supervised, unsupervised, or reinforcement learning tasks.
- Building hybrid quantum-classical ML pipelines that leverage near-term quantum devices (Noisy Intermediate-Scale Quantum, or NISQ) alongside classical hardware.
- Simulating QML models on classical systems to compare performance and identify potential quantum advantages.
- Collaborating with data science teams to identify real-world problems best suited for QML approaches.
- Publishing and presenting research results to foster cross-pollination between quantum computing and AI communities.

Essential Skills

- **Machine Learning Fundamentals:** Familiarity with modern ML frameworks (TensorFlow, PyTorch) and standard models (CNNs, RNNs, Transformers).
- **Quantum Computing:** Knowledge of quantum gates, circuits, and hardware constraints.
- **Hybrid Algorithms:** Ability to design and optimise workflows that combine quantum subroutines with classical pre/post-processing.
- **Mathematical Rigour:** Comfort with linear algebra, optimisation, and probability theory at the graduate level or higher.
- **Adaptability:** Willingness to pivot research directions as quantum hardware evolves and new algorithmic breakthroughs emerge.



Why Join Quantum



Who Should Join Quantum

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